



Review Article

Critical review on plant-derived quorum sensing signaling inhibitors in *Pseudomonas aeruginosa*

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ABSTRACT

Pseudomonas aeruginosa, a biofilm-forming organism with complex quorum mechanisms (*Las*, *Rhl*, *PQS*, and *IQS*), poses an imminent danger to the healthcare sector and renders current treatment options for chemotherapy ineffectual. The pathogen's diverse pathogenicity, antibiotic resistance, and biofilms make it difficult to eradicate it effectively. Quorum sensing, a complex system reliant on cell density, controls *P. aeruginosa*'s pathogenesis. Quorum-sensing genes are key components of *P. aeruginosa*'s pathogenic arsenal, and their expression determines how severe the spread of infection becomes. Over the past ten years, there has been a noticeable increase in the quest for and development of new antimicrobial medications.

Quorum sensing may be an effective treatment for infections triggered by bacteria. Introducing quorum-sensing inhibitors as an anti-virulent strategy might be an intriguing therapeutic method that can be effectively employed along with current medications. Amongst the several speculated processes, a unique anti-virulence strategy using anti-quorum sensing and antibiofilm medications for targeting pseudomonal infections seems to be at the forefront. Due to their noteworthy quorum quenching capabilities, biologically active phytochemicals have become more well-known in the realm of science in this context. Recent research showed how different phytochemical quorum quenching actions affect *P. aeruginosa*'s QS-dependent pathogenicity. This review focuses on the most current data supporting the implementation of plant bio-actives to treat *P. aeruginosa*-associated diseases, as well as the benefits and future recommendations of employing them in anti-virulence therapies as a supplementary drug development approach towards conventional antibiotic approaches.

1. Introduction

The continuous dependency on antibiotics for antimicrobial treatment has resulted in the global rise of isolates of bacteria resistant to

medications [1]. Antimicrobial resistance (AMR) is one of the public health's most serious problems. Antibiotic misuse and overconsumption have led to the evolution of infectious agents into multidrug-resistant (MDR) strains [2]. The ESKAPE infectious agents, including

Abbreviations: AMR, Antimicrobial resistance; MDR, Multidrug-resistant; QS, Quorum sensing; AIs, Autoinducers; QSIs, QS inhibitors; QQ, Quorum quenching; 3O-C12-HSL, AHL N-(3-Oxododecanoyl)-L-homoserine lactone; C4-HSL, N-butyl-L-homoserine lactone; RIs, Rhamnolipids; PCN, Pyocyanin; PVD, Pyoverdine; HCN, Hydrogen cyanide; AprA, Alkaline protease; Sub-MICs, sub-minimum inhibitory concentrations; EPS, Extracellular polymer substances; NLCs, Nanostructured lipid carriers; NPs, Nanoparticles; OSC, Organosulfur compounds; sRNA, Small regulatory RNA; Pch, Pyochelin; SEM, Scanning electron microscopy; PDT, Photodynamic therapy; NC, Natural compounds; MS, Mass spectrometry.

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Enterococcus faecium, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* strains, give rise to considerable difficulties in controlling infection causing illnesses [3]. The United Nations ad-hoc Intergovernmental Advisory Committee on AMR predicts that drug-resistant diseases will cause 10 million fatalities annually by 2050 and lead to catastrophic poverty for 24 million individuals by 2030 [4]. This situation is sadly mirrored in emerging nations like India, where the usage of antibiotics has almost doubled [5]. It is concerning to note that by 2050, India is expected to account for one-fifth of all AMR-related fatalities worldwide [6]. Table 1..

P. aeruginosa is a highly adaptable gram-negative opportunistic bacterium belonging to ESKAPE infectious agents linked to infections in healthcare facilities, often catastrophic for individuals suffering from cystic fibrosis [7]. This pathogen is extremely invading and toxic, with many pathogenic characteristics, including extracellular enzyme release, exotoxins formation and biofilm development [8]. Its high genomic adaptability underpins its remarkable phenotypic versatility and quick establishment of various antibiotic (ATB) resistance pathways [9]. It involves producing enzymes that break down ATBs, such as β -lactamases and

carbapenemases, membrane permeation defaulted owing to porin deficiencies, implementing efflux systems, and modifying pharmacologic targets [10–12]. Moreover, *P. aeruginosa* may form biofilms that increase its inherent drug resistance and pathogenicity. Table 2..

All of these factors work in tandem to manipulate the immune systems of hosts and attack host tissues, which increases fatalities and morbidity in immunity-weakened patients. Since *P. aeruginosa* forms biofilm on healthcare devices and exploits the host with modified microflora due to administering broad-spectrum antibiotics, scientists have been investigating and creating novel treatment approaches to combat these infections [13,14]. The virulence of *P. aeruginosa* is controlled by numerous interconnected processes strictly controlled by the regulated mechanism called quorum sensing (QS) [15,16]. QS is a widely recognized cell-to-cell signal interaction mechanism that enables *P. aeruginosa* to assess their cell density by producing signaling molecules called autoinducers (AIs) capable of coping with environmental and community modifications [17,18]. QS modulates gene expression via signaling molecules, including biological luminescence, pigmentation, biofilm development, and pathogenic secretory factors [19]. Table 3..

Table 1

A list of bioactive plants have been demonstrated for QSI and biofilm inhibition towards *P. aeruginosa* infections.

Plant species	Family	Target pathogen	MIC	Mode of action	References
<i>Amomum tsao-ko</i>	Zingiberaceae	<i>P. aeruginosa</i>	2 mg/mL	Suppressing the development of biofilms by QS response modulation	[86]
<i>Andrographis paniculata</i>	Acanthaceae	<i>P. aeruginosa</i>	5 mg/mL	Modulating QS-controlled pathogenicity hinders biofilm formation and interferes with pathogen-triggered Mitogen-activated protein kinase (MAPK) circuit initiation.	[87]
<i>Anogeissus leiocarpus</i> (DC) Guill. et Perr.	Combretaceae	<i>P. aeruginosa</i> PAO1	1.25 mg/mL	Inhibition of QS regulator genes	[88]
<i>Cornus controversa</i>	Cornaceae	<i>P. aeruginosa</i> PAO1	>2%	Anti-biofilm and QS inhibitive characteristics	[89]
<i>Cassia alata</i> L.	Caesalpiniaceae	<i>P. aeruginosa</i>		Interfering with QS-associated infectious factors and suppression of bacterial mobility	[90]
<i>Mangifera indica</i>	Anacardiaceae	<i>P. aeruginosa</i> PAO1	>0.8 mg/mL	Concentration-based suppression of QS-assisted virulence determinants and biofilm development	[91]
<i>Laserpitium ochridanum</i>	Apiaceae	<i>P. aeruginosa</i>	0.5 mg/mL	Reduced biofilm growth and generation of QS-assisted factors that promote pathogenesis	[92]
<i>Syzygium jambos</i> (L.) Alston	Myrtaceae	<i>P. aeruginosa</i> PAO1	1 mg/mL	QS suppression and biofilm disintegration	[93]
<i>Syzygium cumini</i> (L.) ethyl acetate fraction	Myrtaceae	<i>P. aeruginosa</i>		The inhibitory effect of QS controlling pathogenicity and biofilm development	[94]
<i>Parkia javanica</i>	Fabaceae	<i>P. aeruginosa</i>	180 μ g/mL	Reduced swarming, virulent factor release, and biofilm growth by QS inhibition	[95]
<i>Pistacia atlantica</i>	Anacardiaceae	<i>P. aeruginosa</i>	0.5 mg/mL	Suppression of QS-controlled virulence by targeting transcription controller, LasR	[96]
<i>Solanum torvum</i>	Solanaceae	<i>P. aeruginosa</i>		Reduced bacterial pathogenicity at sublethal levels	[97]
<i>Psoralea corylifolia</i>	Fabaceae	<i>P. aeruginosa</i>	>1 mg/mL	Hindrance of QS-controlled pathogenic factors by modulation of stable association with transcription regulators, LasR and RhlR	[98]
<i>Terminalia bellirica</i>	Combretaceae	<i>P. aeruginosa</i>	>0.5 mg/mL	Reduced biofilm development by disrupting QS regulating behavior	[99]
<i>Herba patriniae</i>	Caprifoliaceae	<i>P. aeruginosa</i>		Downregulating biofilm-related genes	[100]
<i>Galla chinensis</i>	Anacardiaceae	<i>P. aeruginosa</i> PAO1	2 mg/mL	Inhibiting QS-assisted swarming mobility	[101]
<i>Hypericum perforatum</i>	Hypericaceae	<i>P. aeruginosa</i> PAO1		Downregulating of QS-controlled pathogenic phenotypes by altering QS signaling circuit	[102]
Mosla soochouensis Matsuda	Mosloflavone	<i>P. aeruginosa</i> PAO1	125 μ g/mL	Hindered the PCN production (59 %), LasB elastase (39 %) and chitinase (66 %). Mosloflavone downregulated the expression of QS-controlled genes <i>lasI</i> , <i>lasR</i> , <i>rhlI</i> , <i>chiC</i> , <i>rhlA</i> , <i>rhlR</i> , <i>lasB</i> , <i>phzM</i> , <i>toxA</i> , <i>aprA</i> , <i>exoS</i> , <i>algD</i> and <i>pelA</i> .	[103]
<i>Alstonia scholaris</i>	3, 5, 7-Trihydroxyflavone	<i>P. aeruginosa</i>	100 μ g/mL	The plant-derived component exhibited immense potentiality in inhibiting biofilm and QS-associated pathogenic synthesis in <i>P. aeruginosa</i> .	[104]

Table 2Therapeutical plants and their extracts suitable for QS inhibition in *P. aeruginosa*.

Plant species	Type of extracts	Target pathogen	Mode of action	References
<i>Angelica panicii</i> Vandas	Liquid, ethanolic and methanolic	<i>P. aeruginosa</i> PAO1	Inhibiting PCN synthesis, biofilm growth, twitching and flagellar mobility	[105]
<i>Camellia nitidissima</i> C.W. Chi	Dichloromethane fraction	<i>P. aeruginosa</i> PAO1	Hindering PCN synthesis, swarming and swimming locomotion	[106]
<i>Echinophora sibthorpiana</i> Guss.	Liquid, ethanolic and methanolic	<i>P. aeruginosa</i> PAO1	Suppressing PCN synthesis, biofilm creation, twitching and flagellar movement	[107]
<i>Mangifera indica</i> L.	Acetone Ethyl acetate Methanolic	<i>P. aeruginosa</i> PAO1	Impeding violacein and PCN formation, reduction of elastase, protease, chitinase EPS synthesis, swarming mobility and biofilm creation	[108]
<i>Micromeria thymifolia</i> (Scop.) Fritsch	Essential oil	<i>P. aeruginosa</i> PAO1	Interfering PCN synthesis, biofilm creation Twitching and flagellar movability	[109]
<i>Myrmecodia tuberosa</i> Jack	Ethanolic Ethyl acetate Hexane fraction Liquid Methanolic	<i>P. aeruginosa</i> PAO1	Impeding swarming, swimming, twitching mobility and biofilm growth	[110]
<i>Rumex alveolatus</i> Los-Losinsk	Methanolic	<i>P. aeruginosa</i> ATCC 1310	Hindering PCN formation	[111]
<i>Trapa natans</i> L.	Acetone Ethyl Acetate Methanolic	<i>P. aeruginosa</i> PAO1	Suppressing biofilm creation, swarming mobility, PCN, violacein and elastase synthesis	[112]
<i>Armillaria mellea</i> (Vahl: Fr.) Kummer	Methanolic	<i>P. aeruginosa</i> PAO1	Inhibitory effects on LasR and PqsR system Interfering biofilm development	[113]
<i>Melilotus albus</i> Medik	Acetone Ethanolic Ethyl acetate	<i>P. aeruginosa</i> ATCC 27,853	Inhibition of biofilm creation	[114]

Signaling molecules, the central component of the QS system, are essential in transferring signals. It has been identified that inhibiting the QS mechanism of bacteria can dramatically lower the infectious potential of pathogenic agents without negatively impacting the pathogen's ability to proliferate [20,21]. Researchers in the microbiology domain are fascinated by this bacterial communicating mechanism, which plays a role in both acute and long-term infections. Thus, harnessing the QS system as a therapeutic target for bacterial illnesses offers a fresh approach to effectively halting the emergence of bacterial resistance [22]. It comprises QS inhibitors (QSIs), which either directly associate with AIs to restrict the acquisition of signals or target signal output to accelerate the breakdown of AIs, safeguarding anti-QS response known as quorum quenching (QQ) [23]. While QS functions as an accessory component in regulating virulence factors in *P. aeruginosa*, its suppression with QSIs does not impose selective-pressure on the development of anthropogenous resistance, which is often observed following the treatment with traditional antibiotics [24,25].

Moreover, one intriguing feature of QSIs is their ability to weaken a specific pathogen's QS pathways at doses below the threshold without endangering life or causing survivability stress [26]. This approach of QSIs has the potential for treating pseudomonal infestations due to its anti-virulence properties. Many conventional plant-derived natural products and their phytochemicals have been shown to have strong QQ potential toward *P. aeruginosa*, and they are therefore regarded to be risk-free for therapeutic usage [27]. Secondary metabolic products of plants and biologically active phytochemicals have recently acquired popularity among scientists. Studies on conventional remedies have demonstrated to be lucrative for pharma companies because plants contain a variety of secondary metabolites, including alkaloids, flavonoids, polyphenols, saponins, tannins and terpenoids, which have been demonstrated to both strengthen the immune system and combat infection-causing agents [28–30].

Plant's antibacterial potency has shown to be unfathomable, showing effectiveness even at very low dosages. In addition to their inherent occurrence, ease of access, cost-effective, potent pharmaceutical characteristics, little toxicity to cells, and low likelihood of resistance development, in comparison with antibiotics, plants are known to possess antibacterial capabilities across the globe [31–33]. Naturally active phytochemicals are shown to have an enduring and intriguing potential in the healthcare sector as therapeutical agents to suppress the pathogenic mechanism in *P. aeruginosa* [34]. In light of the phenomenal pharmaceutical significance of phytochemicals, this review attempts to give an overview of the recent advancements by correlating the anti-virulent characteristics of plant biological agents to the inhibition of QS and pathogenicity in *P. aeruginosa*. These strategies are highly useful for reviving to enamel medicines and preventing the emergence of resistance in *P. aeruginosa* against medications, which will benefit humanity and the health care system.

2. Quorum sensing system of *P. Aeruginosa*

In bacterium, cell-to-cell interaction promotes synchronized responses that enable the expression of several pathogenic features that cannot be achievable for individual cells. The signaling system in *P. aeruginosa* is one of the most complicated and extensively investigated QS mechanisms. *P. aeruginosa* is an extremely dangerous and well-known human infectious agent because its multiplexed signaling pathways work together to control bacterial infectiousness [35]. The QS circuit in *P. aeruginosa* is represented by a universal regulatory mechanism under the direct or indirect control of over 10 % of its genomics and around 20 % of the bacterium proteomics [36]. QS in *P. aeruginosa* is regulated by three major mechanisms: Las, Rhl, and Pqs, as shown in Fig. 1.

Table 3

Virulence-inhibiting therapeutics sourced from plant-based products and their potential in attenuating QS and biofilm formation in *P.aeruginosa*.

Chemical class	Phytochemicals and source	Target pathogen	MIC	Mode of action	References
Alkaloids	Berberine (<i>Berberis</i> sp.)	<i>P. aeruginosa</i> PAO1	1.25 mg/mL	Targeted RhlR of QS pathway and regulated biofilm mechanisms	[248]
	Hordenine (Fig. 2.e) (<i>Hordeum vulgare</i>)	<i>P. aeruginosa</i> PAO1	2.5 mg/mL	Inhibition of QS controlling genes and biofilm creation	[249]
	Reserpine (<i>Rauwolfia serpentina</i>)	<i>P. aeruginosa</i> PAO1	0.8 mg/mL	Impeding of QS-assisted virulent phenotypes, biofilm disintegration and mobility	[250]
	Caffeine (Fig. 2.k) <i>Coffea arabica</i>	<i>P. aeruginosa</i> (MTCC424)	200 µg/mL	Breakdown of biofilm by suppressing QS mechanism while maintaining the survival of cells	[251]
Anthocyanins	Proanthocyanidins (<i>Vaccinium macrocarpon</i> L.)	<i>P. aeruginosa</i> PA14		Blocking of QS-regulated virulent factors.	[252]
	Anthocyanidins	<i>P. aeruginosa</i> PAO1	0.45 mg/mL	Biofilm breakdown	[253]
	Pyrananthocyanins	<i>P. aeruginosa</i>	> 512 µg/mL	Interference with the transcription of numerous QS-associated genes in the <i>P. aeruginosa</i>	[254]
Flavonoids and polyphenols	Baicalein (<i>Scutellaria baicalensis</i>)	<i>P. aeruginosa</i>	0.256 mg/mL	Reduces QS-induced pathogenicity and breaks biofilms	[255]
	Quercetin glucopyranoside (<i>Allium cepa</i>)	<i>P. aeruginosa</i>	0.4 mg/mL	Modulating QS-controlled response	[256]
	Calycopterin (<i>Marcetia latifolia</i>)	<i>P. aeruginosa</i> (ATCC27853)	34 µM	The suppression of QS-assisted PCN synthesis and swarming mobility	[257]
	Resveratrol	<i>P. aeruginosa</i>	>400 µM	Impeding QS-controlled phenotypes by reducing oxidative stresses	[258]
				Adhere to the QS functional sites	
	Catechin	<i>P. aeruginosa</i>	128 µg/mL	Reduced expression of QS-associated genes	[181]
	Epigallocatechin gallate (EGCG)	<i>P. aeruginosa</i>	200 µg/mL	Curcumin at a sub-MIC 200 µg/mL caused a 21 % reduction in the <i>P. aeruginosa</i> PA14 strain's ability to produce 3-oxo-C12-HSL.	[259]
	Curcumin	<i>P. aeruginosa</i> PA14			[260]
	Cinnamic acid (<i>Averrhoa carambola</i>)	<i>P. aeruginosa</i>	0.5 mg/mL	Impaired QS-regulated pathogenicity	[261]
	Cinnamaldehyde (<i>Cinnamomum</i> sp.)	<i>P. aeruginosa</i>	>11.8 mM	Regulates c-di-GMP signaling and causes disruptions in biofilms	[190]
Phenylpropanoids	Chlorogenic acid (Fig. 2.s) (<i>Camellia sinensis</i>)	<i>P. aeruginosa</i>	>5 mg/mL	Hinders QS-controlled infectious factors and biofilm development	[262]
	Ferulic acid	<i>P. aeruginosa</i> PAO1	1 mg/mL	Regulating EPS synthesis may impede biofilm creation	[263]
	Diallyl sulfide (<i>Allium sativum</i>)	<i>P. aeruginosa</i> PAO1		Diallyl sulfide suppressed transcription of QS system genes, comprising <i>lasR</i> , <i>rhlI/rhlR</i> and <i>pqsABCDE/pqsR</i> ; hence, synthesis of the signaling molecules C4-HSL and PQS was impeded.	[264]
	Diallyl trisulfide (<i>Allium sativum</i>)	<i>P. aeruginosa</i> PAO1		Diallyl trisulfide inhibited the transcription of genes related to PAO1 QS systems along with virulent genes, such as exoprotease, elastase, lectin, and RLs, in PQS, PCN production and biofilm creation.	[265]
Terpene and Terpenoids	Sulforaphane	<i>P. aeruginosa</i>	100 µM	Prevented biofilm growth and pyocyanin generation. Attachment to LasR and block its activation.	[266]
	Tormenteric acid congener (<i>Callistemon viminalis</i>)	<i>P. aeruginosa</i>	25 µg/mL	Biofilm breakdown via reducing the discharge of eDNA and capsule polysaccharides derived from biofilms	[267]
	Betulin & betulinic acid (<i>Syzygium jambos</i>)	<i>P. aeruginosa</i>	250 µg/mL	Interfering with QS-triggered pathogenicity and biofilm structure	[268]
	Carvacrol (<i>Origanum vulgare</i>)	<i>P. aeruginosa</i>	>3.9 mM	Targeting LasI and modifies QS-regulated behavior	[220]
	Phytol (Plant chlorophyll)	<i>P. aeruginosa</i>	19 µg/mL	Changes in the bacterial mobility characteristics	[269]
	Glycyrrhetic acid (<i>Glycyrrhiza glabra</i>)	<i>P. aeruginosa</i>	160 µg/mL	Prevents the production of biofilms and the release of infectious factors	[270]
	Tormenteric acid	<i>P. aeruginosa</i>	25 µg/mL	Reduced the expression levels of <i>rhlI</i> , <i>rhlR</i> , <i>LasR</i> , <i>lasI</i> , and <i>lasB</i>	[271]
	Carvacrol	<i>P. aeruginosa</i>	70.4 mg/mL	Carvacrol hindered the growth of biofilms by 88 % for <i>P. aeruginosa</i> and reduced the enzymatic action of the biofilm by 40 %-100 %	[218]
	Carvacrol combined with pepsin/trypsin	<i>P. aeruginosa</i>	5 mg/mL	When combined with trypsin or pepsin in addition to carvacrol, the <i>P. aeruginosa</i> biofilm community was reduced more than when carvacrol was used itself	[272]
	Sabinene (<i>Zingiber Cassumunar</i>) EOs	<i>P. aeruginosa</i>	62.5 µg/mL	Substantial impedance of biofilm creation and QS mechanism, along with reduced levels of PCN, PVD and proteolytic action	[273]
	Carvone and limonene (Caraway)	<i>P. aeruginosa</i>		Lowered synthesis of virulent associated factors controlled by QS mechanisms in Gram-ve bacterium	[274]
	Piperine	<i>P. aeruginosa</i>	16 µg/mL	Inhibition of biofilm creation and reduced expression of QS gene <i>lasI</i>	[275]

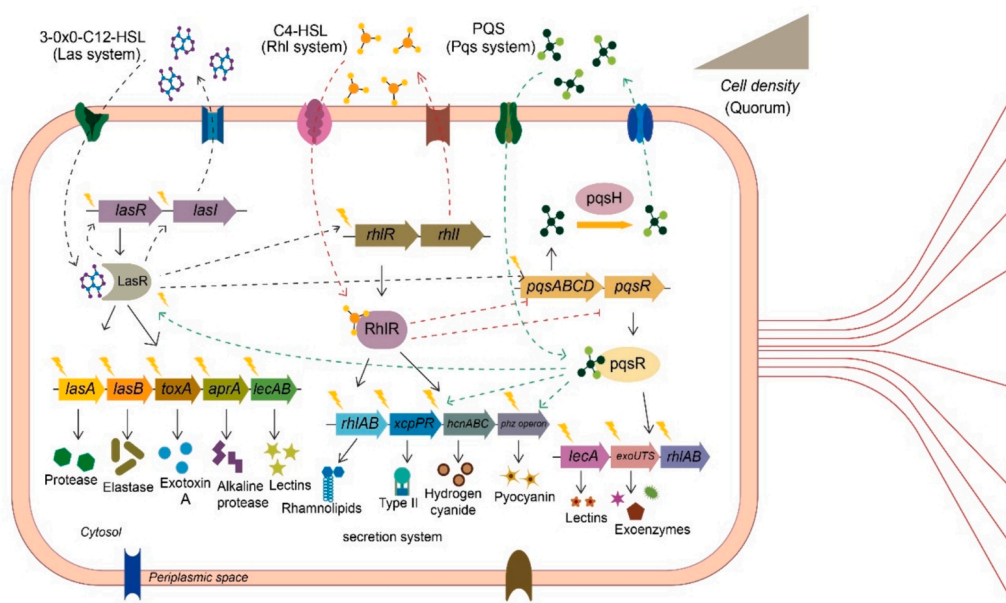


Fig.1. QS circuit of *P. aeruginosa*. Las, Rhl, and Pqs subsystems govern complicated signaling within cells through their associated QS signaling molecules 3-oxo-C12-HSL, C4-HSL, and PQS. Inducing these QS systems increases the infectiousness of *P. aeruginosa*, leading to the amplification of the disease and the seriousness of the infection [37].

2.1. Las system

The Las system consists of two crucial components, LasR and LasI, which function as transcriptional regulators and synthase proteins [38]. LasI synthesizes the AHL N-(3-Oxododecanoyl)-L-homoserine lactone (3O-C12-HSL). 3O-C12-HSL is a required molecule for initiating QS in *P. aeruginosa* via its interaction with LasR to create proteins that contain numerous monomers and induce as a transcriptional factor [39,40]. LasR protein with numerous monomers can bind to specific genes and regulate transcription. The system regulates the expression of various pathogenic factors associated with the pathogenic phase of acute diseases, including exotoxin A and Las(A, B) elastases [41].

The Las circuit positively modulates gene expression of the LasI/R and PQS systems that code synthase genes (*rhlI*, *pqsH*) and quorum sensing signaling molecule receptors (*rhlR*, *pqsR*) [42].

2.2. Rhl system

RhlI/RhlR is the second type of strategy employed by *P. aeruginosa*. The synthase enzyme RhlI is responsible for initiating AHL N-butyl-L-homoserine lactone (C4-HSL) synthesis [43]. C4-HSL binds to RhlR as a transcriptional controller, resulting in a complex system that regulates the expression of many genes [44]. Hence, 3O-C12-HSL/C4-HSL is produced by *P. aeruginosa* as the signaling molecule. The rhamnolipids (RLs) and proteins that generate unwanted bacterial proteins within *P. aeruginosa* in the host cytoplasm are regulated by the RhlI/RhlR mechanism [45]. The Rhl QS system controls biofilm growth and regulates RLs. It is a glycolipid biologic surfactant of rhamnose and 3-(hydroxyalkanoyloxy)alkanoic acid [46]. They enhance motility, prevent the bacterial biofilm from attaching rigidly to the substrate's surface during initial interactions, and maintain the fluid channels for transporting H₂O, O₂, and nutrients to the biofilm substrate [47].

2.3. PQS system

The PQS system, also known as the third QS system of *P. aeruginosa*, is associated with the formation of 2-heptyl-3-hydroxy-4-quinolone through the use of the exterior membrane vesicular and the PQS ABCDE gene as a signaling molecule [48]. Moreover, the PQS affects the

synthesis of RLs and additional constituents linked to the development of biofilms [49]. Previous research reported that LasR positively regulates RhlR and PQS, whereas RhlR negatively regulates the PQS system [50]. QS in *P. aeruginosa* modulates virulent components such as RLs, pyocyanin (PCN), and pyoverdine (PVD). RLs are a kind of glycolipids that are best described as bacterial surfactants. They include a glycosyl head group, rhamnose, and a 3-(hydroxyalkanoyloxy) alkanolic fatty acid at the end of the chain [51].

RLs are the only pathogenic agents that cause weaknesses in people with ventilator-associated infestations. In *P. aeruginosa*, QS controls two major luminous pigments: PCN and PVD. PCN is a blue redox-active secondary metabolite that appears as a greenish-blue pigment called blue pus, signifying the presence of *P. aeruginosa* growth in the host organism [52]. A significant quantity of PCN has been detected in people with the condition cystic fibrosis and severe otitis media, which interferes with the capacity of cells to function normally [53]. PCN is generated when the level of carbon or O₂ decreases due to nutrition deprivation or the exponential development phase. As the bacteria's multiplication rate decreases, PCN production increases considerably. It is a zwitter ion with two functioning groups that may readily invade the host cell membrane [54].

PVD is an extracellular, luminous molecule of yellowish-green color. Besides being a Fe transporter, PVD is an intermediary molecule in synthesizing two extracellular pathogenic components: endoprotease and lethal exotoxin A [55]. PVD has been involved in generating resilient and established biofilms in *P. aeruginosa*. Type III quorum sensing systems and type IV pili are linked to more severe infections in the host organism. During mammalian host encroachment, *P. aeruginosa* produces numerous pathogenic components to overcome inherent barriers involving the host's immune cells. According to reports, *P. aeruginosa* cannot cause infestations unless it possesses type III secretory systems, type IV and QS [56,57].

2.4. Release of pathogenic factors

P. aeruginosa produces a wide range of infectious factors associated with the host's invasive mechanism, comprising adhesins and lectins, proteolytic enzymes and Fe transporters, which facilitate tissue encroachment exotoxins, distinct exoenzymes released by intricate

secretory mechanisms, RLs, and hydrogen cyanide (HCN), which maintains bacterial persistence by inhibiting the immune system [58,59]. *P. aeruginosa* has five distinct secretory pathways.

Type I and type V secretion systems release toxicants from the cell's outer setting, such as alkaline protease (AprA) and extracellular adhesion. Type V secretion mechanisms use a two-stage secretory process that transmits molecules across the periplasm [60]. The most important secretory mechanism, Type III secretion, instantly infuses exoenzymes Exo(S, T, U, Y) within the nuclear region of receptive immune cells, causing them to die. Finally, the type II secretion system produces pathogenic agents such as exotoxin A, elastases A and B, type IV proteases, and AprA [61]. Certain QS-regulated genes that influence physiological activities, pathogenic features, and the dispersal of pathogenic drivers affecting the human host include *lasA*, *lasB*, *aprA*, *hcnABC*, *lecA*, *phzABCDEFG*, *phzM*, *rhlAB*, and *toxA* [62]. When the *lasB* gene is activated, elastases are generated, which increases the host's tissue disintegration (elastin, collagen) and *P. aeruginosa*'s iron uptake from degraded host proteins [63].

In comparison, QS control of the *lasA* gene is crucial for protease production, contributing to *P. aeruginosa*'s host entrance by breaking down the host's epithelium barrier [64]. *AprA*, *rhlAB*, and *toxA* genes are essential for synthesizing exotoxin A, alkaline protease, and rhamnolipids in *P. aeruginosa*, resulting in infection persistence, immune system encroachment, and biofilm establishment. It induces the death of cells, cytokine and complementary pathway deterioration, and the mortality of the host's polymorphonuclear lymphocytes and phagocytes [65]. The genes *lecA*, *hcnABC*, and *phzABCDEFG* play a role in the discharge of lectin, HCN, and PCN, which result in ciliary retardation, cellular respiratory failure, and stress owing to oxidation [66].

2.5. Integration and cross-talk

Researchers have proposed various hypotheses to explain the cross-talk among the many QS signaling circuit branches over time. According to Lee and Zhang [67], *P. aeruginosa*'s three QS mechanisms are linked, with the Las system playing a vital role in controlling signaling. It has been demonstrated to control both Rhl and Pqs systems. It is noteworthy to note that a circular paradigm describing the associations amongst *P. aeruginosa*'s QS systems has been proposed by Maura et al. [68]. MvfR (PqsR) regulates the expression of *lasR* and *rhlR* in initial bacterial development, whereas RhlR suppresses the MvfR (PqsR) QS mechanism during post-growth and static stages. The association of MvfR to *lasR* promotes *mvfR* expression and PqsH-aided transformation of HHQ into PQS, leading to increased MvfR functionality. It was demonstrated that these processes operated as strict autoregulatory circuits that controlled *P. aeruginosa*'s QS network. However, QSSMs are chemical compounds that *P. aeruginosa*'s QS systems use to interact with each other. The LasR-3-oxo-C12-HSL complexes promote the expression of *rhlR*, *rhlI*, *lasI*, and other pseudomonal pathogenic genes, forming a favorable feedback circuit [69].

An additional positive feedback looping is produced by the RhlR-C4-HSL complex, which results in dimer formation and promotes the production of RhlI and its regulon [70]. The LasR-3-oxo-C12-HSL complexes control PqsR, an expression controller of the *pqsABCD* operon. It promotes the production of PQS and the expression of *pqsH*, a gene that may encode a monooxygenase [71]. Furthermore, it has been discovered that PQS improves *rhlI* transcription, which affects C4-HSL production and the overall effect of the Rhl system [72]. Additionally, RhlR-C4-HSL inhibits the transcription of *pqsR* and *pqs* operon, indicating that the proportion of level among 3-oxo-C12-HSL and C4-HSL is critical for the supremacy of the Pqs signaling system [73]. The Las and Pqs systems regulate the Rhl mechanism; however, the RhlR-C4-HSL complex primarily triggers QS-dependent factors associated with virulence.

These mechanisms act as a chain of events process, directing *P. aeruginosa*'s pathogenicity once quorum is reached. Microbiologists have focused on this remarkable bacterial interaction mechanism due to its

role in short-term and long-term human infections [74]. The inhibition of the QS process, which plays an integral part in microbial infectiousness, is an alternative technique for controlling nosocomial illnesses caused by biofilms. Anti-QS medicines, both manufactured and natural, effectively treat *P. aeruginosa* infestations by lowering biofilm thicknesses and virulence [75].

QS attenuation involves two approaches: signal deterioration and signal mimicking. These processes impede genetic regulatory systems and impair genes involved in pathogenicity and biofilm formation [76]. QS/biofilm antagonists have been shown to have minimal selective stress on pathogens to evolve resistant. These treatments aim to reduce the development of virulent factors and biofilms while not influencing the development of pathogens. Finding and examining non-toxic chemicals that limit bacterial interactions and development is necessary to tackle biofilm-associated illnesses. Understanding the molecular processes and possible targets of these communication networks can aid in creating innovative antimicrobial medicines for selective and defensive treatment.

3. Exploring plant-based phytochemicals as therapeutics towards *P. Aeruginosa*

With enhanced molecular technologies, researchers worldwide are exploring the possibility of developing innovative antimicrobials. Recent research has shown that biologically active phytochemicals can effectively target *P. aeruginosa* by inhibiting its QS, reducing factors associated with virulence and hindering biofilm development [77,78]. The chemical architecture of biologically active phytochemicals that have been demonstrated to have anti-pathogenic, anti-biofilm, and quorum quenching (QQ) properties toward *P. aeruginosa*, as shown in Fig. 2. The heterocyclic ring arrangement in most QQ phytochemicals is comparable to the configuration of AHL molecules.

This architectural confirmation may develop the capacity to attenuate signal receptors and establish feasible persistent associations with QS receptors [79,80]. Ajoene, allicin, and curcumin molecular frameworks resemble AHL side chains but possess differing oxygenation stages. The lactone ring is substituted by an aromatic component in carvacrol, cinnamaldehyde, eugenol, 6-gingerol and zingerone, making it difficult to open them up [81]. Baicalein, naringenin, naringin, quercetin and vitexin (Fig. 2.v) also exhibit a comparable pattern because they all have a complicated polycyclic framework. Furthermore, computer-simulated studies have demonstrated that these phytochemicals have a considerable affinity for the various *P. aeruginosa* QS receptors [82–84].

Nevertheless, since phytochemicals with bioactive properties are not protein-based, no association studies have been able to verify this. However, results from molecular docking tests are favorably associated with the downregulating of critical QS genes in several in vitro and in vivo investigations [84,85]. Hence, plant-based bioactives have been fabricated as versatile substances with no limitations on their application as cutting-edge anti-virulence treatments.

3.1. Phenolic compounds

Flavonoids (derived from the Latin term “*flavus*,” which means yellow color) are diversified plant-based secondary metabolic products with varying phenolic frameworks [115]. On a chemical basis, these compounds share a 15-carbon scaffold with two attached phenyl and heterocyclic rings and an additional phenyl ring. Bananas, blueberries, citrus fruits, onions and parsley contain the largest levels of flavonoids. Citrus extracts contain flavonoids such as kaempferol, quercetin, naringenin, sinensetin and apigenin, which suppress *P. aeruginosa*'s QS-controlled bioluminescence [116].

Polyphenols are classified as phenolic acids, flavonoids, or non-flavonoids based on their chemical makeup, consisting of a benzene ring with a minimum of one hydroxyl group (–OH) attached, as shown in

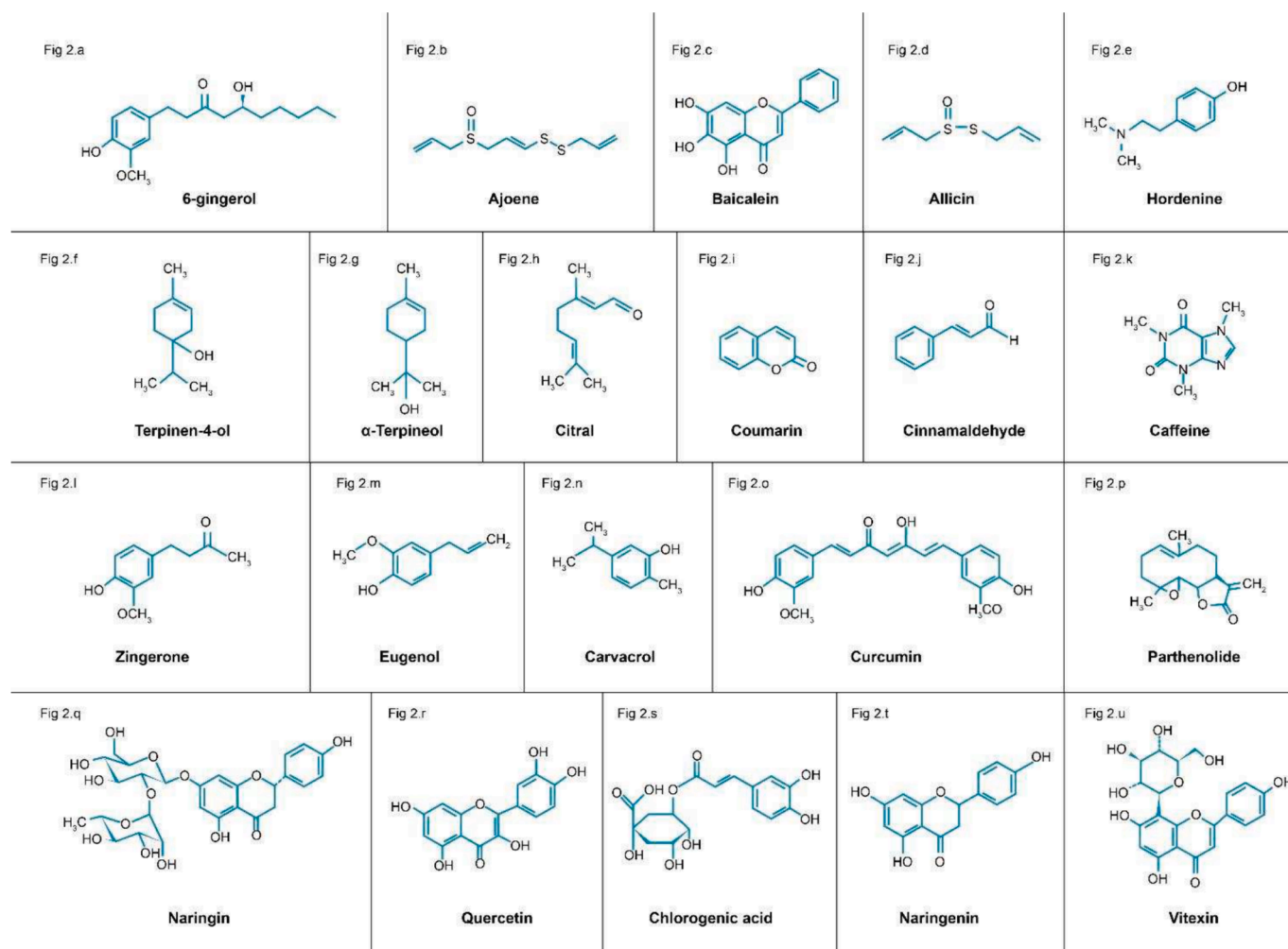


Fig. 2. Chemical frameworks of plant phytochemicals exhibiting QQ and anti-virulence activities towards *P. aeruginosa*.

Fig. 3.

3.1.1. Quercetin

The pericarps of seeds, fruits, and green veggies contain quercetin, a pentahydroxyflavone with potent anti-inflammatory, anti-oxidative, and cancer-fighting properties. Early studies revealed that quercetin might halt *P. aeruginosa*'s mobility and inhibit biofilm development by 95 % at sub-minimum inhibitory concentrations (sub-MICs) [117].

Quercetin (Fig. 2.r) has been shown to have a detrimental effect on pseudomonal biofilms when combined with some antibiotics at sub-MIC levels [118]. Numerous investigations have highlighted quercetin's QQ capability, inhibiting *P. aeruginosa* pathogenic factors. A significant study found that quercetin inhibited biofilm development by 50 % and greatly increased the synthesis of elastase, protease, and PCN without inhibiting bacterial development [119]. Interestingly, important QS genes such as *lasI*, *lasR*, *rhlI*, and *rhlR* were significantly suppressed by quercetin at 16 µg/ml by 35 %, 68 %, 56 %, and 50 %. Hence, this established the groundwork for investigating quercetin as an intriguing treatment. Quercetin at a sub-MIC level of 40 µg/ml hindered alginate synthesis by 65 %, bacterial biofilms by 80 % and extracellular polymer substance (EPS) matrix by 73 % [120].

Significant associations were found when quercetin and the LasR receptor were molecularly docked, potentially eliminating QS by modifying receptor conformational arrangement. Quecan et al. [72] investigated the QQ capability of quercetin aglycone and quercetin 3-β-D-glucoside towards *P. aeruginosa*. The biologically active substances reduced swarming mobility and violacein synthesis in vitro but did not

influence biofilm creation. In silico investigation indicates that the two quercetins effectively interact with the AHL-adhering region of PAO1's LasR receptor, explaining their anti-virulence properties. Quercetin may prevent biofilm growth in *P. aeruginosa* by inhibiting *vfr* and biofilm-associated genes (*pelA*, *pslA*) without affecting QS genes [73]. It was hypothesized to be an outcome of quercetin-induced *lasI*/*lasR* suppression, which altered the *vfr* expression of genes and eliminated pseudomonal biofilm. Additional in silico studies indicate that quercetin can act as a competing inhibitor for the QS signaling molecule when positioned in the AHL-binding sites of *P. aeruginosa*'s LasR protein [121]. Therefore, quercetin is an intriguing choice that has to be carefully assessed in clinical trials on animals before being used in the anti-virulence therapeutic towards *P. aeruginosa*.

3.1.2. Naringenin and naringin

Naringenin (Fig. 2.t) is a colorless flavone, a vital biologically active constituent in citrus fruits, especially grapefruit. Naringin, an O-glycosylated precursor of naringenin, is a further significant phytochemical that gives citrus fruits their bitter flavor. It has been found that both of these flavones possess QQ activity toward *P. aeruginosa* [122]. It was initially shown that naringenin inhibited *P. aeruginosa*'s synthesis of QS-regulated virulent components [123]. In the existence of naringenin, the synthesis of PCN and elastase was suppressed by 87 % and 46 %. This decrease was caused by downregulating the expression of *lasI*, *lasR*, *rhlI*, *rhlR*, and *lasB*, which resulted in a dramatic decline in quorum sensing signaling molecules (QSSM), C4-HSL, and C12-HSL synthesis in *P. aeruginosa* cells administered naringenin in vitro studies. Naringenin

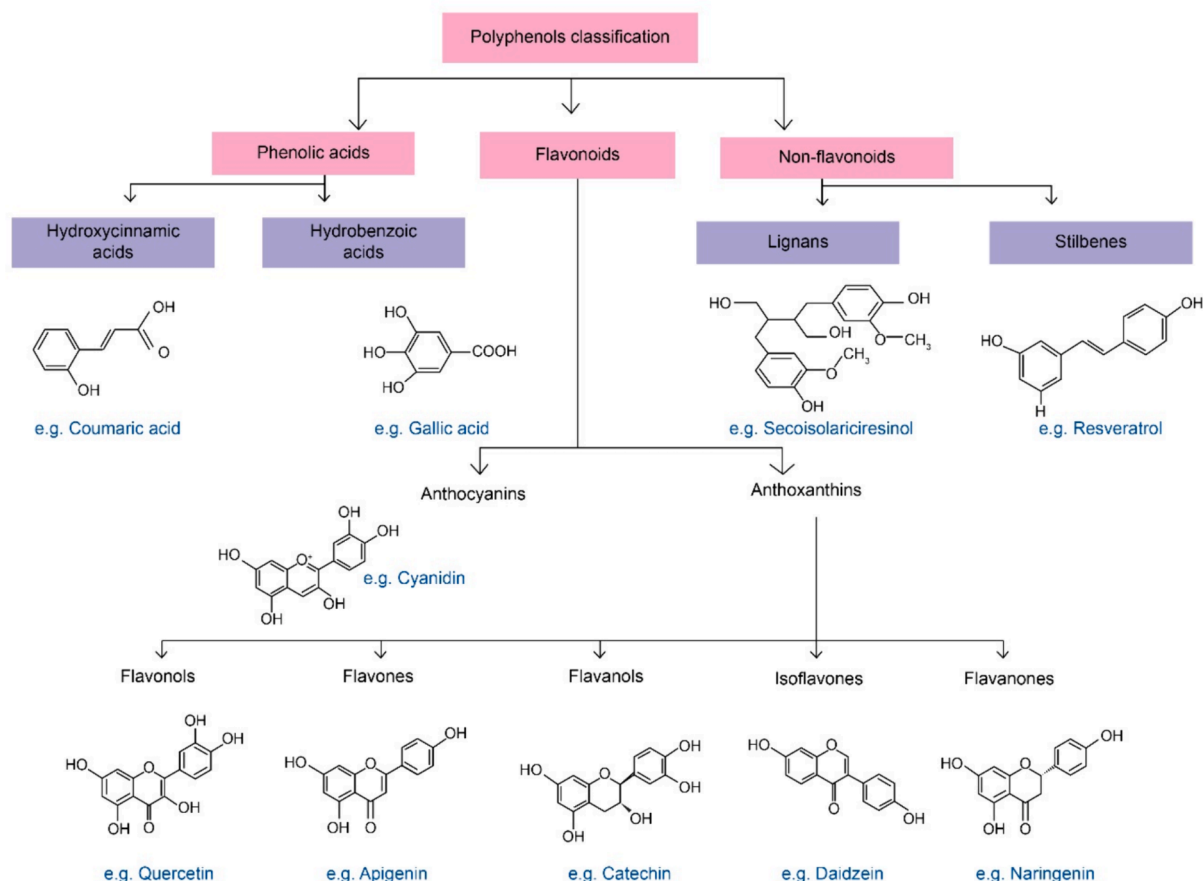


Fig. 3. Classification of Polyphenols.

inhibited QS not only by reducing QSSMs but also by disrupting the RhlR-C4-HSL complex. In a time-based approach, naringenin outperformed C12-HSL for adherence to its corresponding LasR receptor [124].

Naringenin inhibited LasR more effectively when it had a direct relationship with the newly formed receptor form rather than the functionalized version. Naringenin inhibited the production of QS genes and QS-regulated virulence-associated genes during initial bacterial development but diminished its inhibitory impact throughout the stagnant state. Hence, bacterial development influences the competitiveness between C12-HSL and naringenin for the LasR receptor [125]. Naringin (Fig. 2.q) has been demonstrated to successfully eliminate pseudomonal biofilms on inanimate and medical catheter surfaces, hindering swimming locomotion and the development of the EPS matrix [126]. On the other hand, the antibiofilm and anti-virulence effectiveness of naringin towards *P. aeruginosa* was significantly increased when ciprofloxacin and tetracycline were added to the therapy, as contrasted with when they were employed on their own. According to these findings, naringenin and naringenin are possible QQ and anti-biofilm therapeutics for *P. aeruginosa* [127].

3.1.3. Baicalein

Baicalein (Fig. 2.c), a naturally occurring flavonoid found in the rhizomes of *Scutellaria baicalensis*, has antibacterial, antibiotic, and antiviral properties [128,129]. Baicalein inhibited adherence of cells and biofilm biomass formation at sub-MIC levels in *P. aeruginosa*. The expression of QS controlling genes (*lasI*, *lasR*, *rhlI*, *rhlR*, *pqsR*, and *pqsA*) was suppressed, resulting in a decrease in QS signaling molecules such as 3-oxo-C12-HSL and C4-HSL. The study revealed that baicalein effectively inhibited QS signaling and biofilm creation, making it a potential

QS antagonist and antibiofilm agent for infection caused by *P. aeruginosa* [130]. In a *Drosophila melanogaster* (*Drosophila*) infection paradigm, baicalin therapy reduced the infectiousness of *P. aeruginosa*, reduced lung pathological degree of severity, and hastened the elimination of bacteria [131]. Baicalein at sub-MIC levels has been shown to suppress the QS system in *P. aeruginosa*. The formation of QSSMs, C4-HSL, and C12-HSL was reduced by two times, with considerable downregulating of *las* and *rhl* QS genes.

In addition to suppressing MAP kinase and NF- κ B signaling processes, *P. aeruginosa*-affected macrophages produced fewer inflammation-related cytokines IL-1b, IL-6, IL-8, and TNF- α [132]. Hence suggested that baicalein may have anti-virulent characteristics towards *P. aeruginosa*. When baicalein was administered, it triggered inflammatory reactions by stimulating the Th1-driven immunological response, which improved the capacity of the pathogen to be eliminated and increased its anti-QS agent and antibiofilm capabilities. Baicalein-constructed gold nanoparticles (NPs) effectively suppressed QS-regulated pathogenic characteristics in *P. aeruginosa* PAO1, including EPS and swarming mobility. Therefore, the biologically active flavone baicalein was used as a reducer and encapsulation agent to fabricate gold NPs. The findings underlined the potential of phytochemical-based NPs for the medical management of biofilm-associated persistent illnesses [133].

3.1.4. Rutin

Rutin, also called quercetin-3-O-rutinoside, rutoside, sophorin, and vitamin P, is a main flavonoid component in *Citrus sinensis* peel [134]. Rutin, a nutritive component of plant-based foods, has several medicinal benefits such as antioxidative properties, anti-inflammatory, allergic repellent, anti-viral, antibacterial, mutagenic resistant, and superoxide

radical scavenging [135,136]. Rutin can potentially be an intriguing anti-biofilm agent for *P. aeruginosa* because it increases ROS formation in bacteria, which causes oxidative stress and bacterial mortality. When combined with gentamicin, rutin may be an adherence antagonist for managing *P. aeruginosa* infestations. Compounds derived from *Pistacia atlantica* extractions, including rutin, myricetin, 3-O-rutinoside, and kaempferol-3-O-rutinoside, bind strongly to the LasR protein, a key receptor in the QS signaling process. These substances may disrupt normal QS signaling in *P. aeruginosa* [136].

3.1.5. 6-gingerol

Ginger oil, produced from *Zingiber officinale* (ginger), is a phenolic compound that stimulates spicy receptors on the tongue [137]. Kim et al. [138] demonstrated that 6-gingerol had anti-quorum activity towards *P. aeruginosa* employing traditional in vitro investigations. 6-gingerol substantially lowered *lasR* and *rhlR* levels of transcription and RLs synthesis by around 60 % at a dosage of 100 μ M. 6-gingerol (Fig. 2.a), a compound with elongated alkyl chains and *N*-acyl-homoserine lactone components, has been shown to suppress the LasR gene expression. In silico, 6-gingerol has been shown to interact with the QS key controller LasR protein, disrupting the process involved in producing exoproteases, PCN, and *pseudomonas* biofilm. In addition, 6-gingerol inhibited the expression of PQS operons, *phnAB* and *pqsB-E*, T3SS-related genes, QS-regulated ExoT, and elastase [139]. Gingerol analogs disturbed QS functions and reduced biofilm development on RO membranes at a greater rate (37 %) than 6-gingerol (22 %) despite affecting the membrane physically or chemically.

Gingerol analog's increased affinity for the QS protein receptor LasR was also associated with a 78 % reduction in the expression of the *lasB* gene.

3.1.6. Zingerone

Zingerone (Fig. 2.l), also known as vanillylacetone, a biologically active compound found in ginger, contributes to its sweet flavor. Studies have recognized its many pharmacologic qualities as a strong anti-inflammatory, anti-diarrheic, and anti-diabetic medication [140]. However, some researchers have evaluated the potential application of zingerone to inhibit QS processes and prevent *P. aeruginosa* from forming biofilms. Zingerone administration at sub-MICs reduced swimming, swarming, and twitching mobility by over 50 % [141]. Scanning electron microscopy (SEM) showed that zingerone inhibited *P. aeruginosa* biofilm development on catheter surfaces rather than impacting bacterium plankton development.

Kumar et al. [142] demonstrated that Zingerone significantly reduced the synthesis of QSSMs, C12-HSL, C4-HSL, and PQS in clinical strains of *P. aeruginosa* and reference variant PAO1. As a result, the synthesis of pathogenic agents such as elastases, hemolysins, PCN proteases, and RLs was reduced. In silico research evaluated zingerone's QQ capability and affinities for QS receptors. Zingerone interacts strongly with LasR, RhlR, and PqsR receptor attachment sites, inhibiting those QS mechanisms in *P. aeruginosa*. Zingerone-packed chitosan NPs (Z-NPs) exhibited potential anti-virulent capabilities toward *P. aeruginosa*. The Z-NPs were demonstrated to be stabilized under different temperatures, suppressing the synthesis of QSSMs in *P. aeruginosa* employing a reported assay using the *Agrobacterium tumefaciens* A136 strain [143]. Z-NPs decreased bacterial movement by over fourfold, and qRT-PCR tests revealed a significant downregulation of *lasR*, *rhl*, and *rhlR* expression at sub-MIC levels.

Z-NPs demonstrated an anti-biofilm activity, effectively preventing biofilm development and eradicating *P. aeruginosa* biofilms. Z-NPs were shown to be innocuous to HEK-293 cells, with an effective survival rate of around 94 %. They also reduced the infective potential of *P. aeruginosa* and improved the clearance of pathogens from kidney tissues. Z-NPs also increased serum antibacterial properties, expedited the phagocyte absorption and ex vivo apoptosis of *P. aeruginosa* cells by murine phagocytes, and stimulated antibiofilm activity in vitro. These

findings have demonstrated the importance of zingerone as a substitute for antibiotics in treating pseudomonal infestations. Zingerone sourced from *Zingiber officinale* suppresses the synthesis of PCN, elastase, and protease. It also hinders swarming and twitching locomotion in *P. aeruginosa* [144].

3.1.7. Kaempferol

Kaempferol, a naturally occurring flavonoid found in Delphinium, Witch Hazel, grapefruit, and various other plants, has antivirulence properties [136]. The therapeutic herb *Centella asiatica* L. exhibits biological activity in an extract enriched with kaempferol. Kaempferol suppresses *Chromobacterium violaceum* from producing violacein and suppresses the QS-controlled phenotype in a *P. aeruginosa* dose-dependent way. It includes PCN synthesis, elastolytic and proteolytic processes, swarming mobility, and biofilm creation [145].

3.1.8. Eugenol

Eugenol (Fig. 2.m) is a phenolic ingredient in many plant-based products, including aromatic herbs, spices, and oils. Although its wide antibacterial capabilities have been widely recognized for many years, its QQ potential only came to light in the previous decade [146]. Clove oil's major constituent, eugenol, inhibits the transcription of genes regulated by QS [147]. Eugenol inhibits the biofilm population of *P. aeruginosa* by targeting the QS system, which is a key mechanism for biofilm suppression [148–151].

The sub-MICs of methyl eugenol 30 μ g/ml were demonstrated to hinder the flagella-assisted multicellular surface movement by 66 %, swarming mobilities by 24 %, and biofilm development in *P. aeruginosa* strain PAO1 by 90 % [152]. A further investigation found that eugenol suppresses the PAO1 QS mechanism and related virulence-related factors by inhibiting the formation of QSSMs (C12-HSL, C4-HSL) and reducing the expression of *lasR*, *lasI*, *rhlR*, and *rhl* QS genes [83]. In silico studies and in vivo reporter experiments have proven eugenol's competing attachment to the LasR QS receptor, suppressing QS-controlled genes. Eugenol nanoemulsions have been constructed by Lou et al. [153] that have superior QQ effectiveness towards *P. aeruginosa*. These nanoemulsions hinder the synthesis of violacein, PCN, RLs and pseudomonal biofilms at a rate considerably greater than eugenol by itself while suppressing the expression of *lasI* and *rhl* genes, which further reduced the synthesis of 3-oxo-C12-AHL and C4-AHL signaling molecules.

Eugenol inhibited elastase, protease, PCN, and PVD biological synthesis and EPS and RLs formation in the conventional *P. aeruginosa* strain and multi-drug resistant clinical isolates. AgNPs containing eugenol showed a high affinity for QS-related proteins (LasR, LasI, and MvfR) [75]. AgNPs and eugenol together showed a notable affinity for attachment with QS-related proteins. The results showed that the produced NPs could inhibit AHL's generation and the transcription regulator LasR. Additionally, the produced nanoparticle interacts favorably with MvfR, which may lessen *P. aeruginosa*'s QS signaling system's functionality.

3.1.9. Curcumin

Curcumin (Fig. 2.o) is a curcuminoid component found in *Curcuma longa*, a ginger-based therapeutic plant. Curcumin, known for its ability to fight both fungi and bacteria, is widely used in conventional medical practices in India, China, and Southeast Asian countries [154]. Curcumin has been demonstrated to exhibit wide antibacterial activity and strong QQ properties toward *P. aeruginosa*.

Sethupathy et al. [155] reported that curcumin inhibits QS functioning and biofilm formation in *P. aeruginosa* by altering Fe homeostasis and oxidative stress responses. Curcumin affects proteins involved in ROS detoxifying, iron absorption, the tricarboxylic acid cycle, and the formation of PVD and PCN. This inhibitor prevents biofilm creation and regulates the synthesis of protease, elastase, and PCN via QS. A further study demonstrated curcumin as a QS inhibitor that may successfully

prevent biofilm growth, PCN, RLs, and EPS while enhancing the response to antibiotics in *P. aeruginosa* [156]. Curcumin closely resembles AHLs in structure. Curcumin's aromatic ring architecture comprises phenols connected by two (α , β) unsaturated carbonyl groups. Curcumin's phenolic compound mimics the lactone component found in AHL [157–159].

Curcumin binds to both LasR and LuxR via H bonding and hydrophobic associations. This plant-sourced biogenic AHL agonist acts as a QS inhibitor since it may deactivate both proteins through interactions [160,161]. Jadaun et al. [162] investigated the combinatorial impact of curcumin and honey on QS and pathogenic factors in vitro to target *P. aeruginosa*. The synergistic impact reduced the formation of C12- and C4-AHLs, alginate, RLs, hemolysins, PCN, PVD, Pch, LasB elastase, and LasA protease in *P. aeruginosa*. The levels of expression of the *las* and *rhl* QS genes were decreased by about 60 % and 50 %, employing an enzyme-conjugated reporter assay.

Bahari et al. [163] found that administering curcumin at sub-MIC doses throughout the proliferation of bacteria significantly decreased the MICs of azithromycin and gentamicin toward *P. aeruginosa*. Curcumin at sub-MICs and medicines inhibited QS signals, leading to down-regulating *las* and *rhl* genes and reduced PAO1 motility and buildup of biofilm. Zinc-curcumin nanohybrids have been constructed to improve this QQ bioactive's transport and accessibility within *P. aeruginosa* [164].

It had been reported that the nanoformulations suppressed the LasR and RhlR QS mechanisms, safeguarding pulmonary epithelial cells and a mouse affected with PAO1 by reducing the production of pathogenic indicators and developing biofilm without impeding bacterial proliferation. Curcumin-embedded gold NPs on surgical stitches and chitosan-curcumin combinations on stainless-steel bases demonstrated antibacterial activity, stability, and continuous discharge of curcumin [165,166].

Nano-curcumin has greater accessibility and dissolution compared to curcumin.

Nano-curcumin has a higher surface area, leading to better dissolution in water and a longer half-life than curcumin [167–169]. Xue et al. [170] constructed nano-curcumin with increased effectiveness towards *P. aeruginosa* action employing supercritical CO₂ dispersion, outperforming the initially developed curcumin. Sadeghi Mohammadi et al. [171] employed lipid-polymer composite NPs encapsulated with gentamicin and curcumin to suppress the *P. aeruginosa* biofilm population. The study found that NPs containing curcumin outperformed natural curcumin at a comparable quantity due to their higher solubility.

According to Abdulrahman et al. [172], curcumin inhibited *P. aeruginosa* biofilm creation by decreasing EPS production in response to 405 nm laser light exposure. Curcumin-facilitated photodynamic therapy (PDT) resulted in downregulating genes related to QS. Curcumin-triggered antibacterial-PDT inhibits biofilm formation by producing ROS. Reducing the QS mechanism in *P. aeruginosa* may help regulate biofilm growth and eliminate EPS. Curcumin is a highly effective bioactive towards *P. aeruginosa*, making it a promising candidate for antivirulence treatment.

3.1.10. Coumarins

The broad group of plant-derived secondary metabolites known as coumarins (Fig. 2.i) comprise combined benzene and pyrone rings. Some coumarins with antibacterial properties have been identified as phytoalexins, formed in response to an infection from pathogens [173]. Furocoumarins dihydroxybergamottin and bergamottin suppressed *P. aeruginosa* biofilms by 18 % and 27 % while sustaining the development of bacteria [174]. It has been identified that coumarins, aesculetin and aesculin come into contact with the TraR protein's signal-adhering domain and prevent *P. aeruginosa* from forming biofilms [175]. D'Almeida et al. [176] demonstrated the QQ properties of coumarin and its hydroxylation variants towards *P. aeruginosa* in vitro.

Additionally, when exposed to coumarin and its derived compounds,

P. aeruginosa's elastolytic capability decreased. Recently, Zhang et al. [177] have revealed the exact process of *P. aeruginosa*'s coumarin-mediated QQ. Coumarin reduced the expression of significant QS genes such as *lasI*, *rhlI*, *rhlR*, *pqsB*, *pqsC*, *pqsH*, and *ambBCDE*, decreasing QS and suppressing protease and PCN synthesis. Remarkably, coumarin caused cyclic di-GMP signaling to be disrupted by decreasing its levels within cells by 36 % compared to 70 % in its absence, which prevented biofilm development in the in vitro infection model. The combined impact of these coumarin-stimulated processes reduced *P. aeruginosa*'s pathogenicity.

3.1.11. Catechin

Catechins are polyphenolic plant-based chemicals identified in various foods, plants, and fruits. Catechins contain antioxidative, antimicrobial, and tumor growth-reducing characteristics [178]. According to Mombeshora et al. [179], *Triumfetta welwitschii*, an indigenous African therapeutic herb, contains the highest concentration of catechins. This drug reduced the capsular polysaccharide content and eDNA levels of *P. aeruginosa* biofilms. Catechin inhibited biofilm, elastase, PCN, and RL production but did not affect bacterial growth.

The in silico investigation found that catechin-7-xyloside interacts with the QS controller LasR [180]. Another research found 19 naturally occurring substances inhibited *P. aeruginosa*'s QS mechanism. Catechin was shown to be a strong QS inhibitor due to its ability to inhibit LasR activity [181]. Combining catechin and gallic acid predominantly inhibited the transcription of *lasI* and *lasR* in *P. aeruginosa*. The docking investigation demonstrated that catechin was effectively localized within LasR's ligand-binding domain. Catechin forms H bonds with Thr-115 and Ser-129 amino acid residuals associated with the AI ligand. Catechin and gallic acid may inhibit LasR and impair biofilm development in *P. aeruginosa* [182]. Vandeputte et al. [183] discovered that catechin inhibits PCN, elastase, and biofilm growth.

Catechin altered the expression of genes, including *lasB* and *rhlA*, and QS-regulated genes *lasI*, *lasR*, *rhlI*, and *rhlR*. Catechin may interfere with RhlR-based biosensors' identification of the QS signal *N*-butanoyl-L-homoserine lactone. The interference reduces the generation of QS factors. Catechin can effectively limit the biofilm communities of *P. aeruginosa* by interacting with the *las* system, which is the primary QS regulator. Catechin may act as an anti-QS agent, reducing *P. aeruginosa*'s biofilm, infectiousness, and pathogenesis.

3.2. Phenylpropanoids

3.2.1. Cinnamaldehyde

Cinnamaldehyde (Fig. 2.j), a phenylpropanoid produced from Cinnamomum, is a key component of cinnamon oil, contributing to the spice's distinctive fragrance and flavor [184]. Cinnamomum and its derivatives, especially cinnamaldehyde, are being studied for their antimicrobial activities and capacity to prevent the emergence of microbial biofilm.

Cinnamaldehyde successfully eliminates *P. aeruginosa* biofilms via numerous antibacterial mechanisms [185–187]. Cinnamaldehyde suppresses *P. aeruginosa* biofilm development by reducing QS levels. Cinnamaldehyde at sub-MIC (2.2 mM) levels suppresses the *las* and *rhl* mechanisms by decreasing the function of protein regulators LasR and RhlR.

This action lowered extracellular virulent factors such as elastase (22 %), PCN (32 %), and protease (65 %), suppressed the transcription of the RLs gene by 100 %, and hindered the emergence of biofilm in *P. aeruginosa* PAO1 by 26 % [188]. Leoni et al. [189] conducted a detailed analysis of the QQ capability of cinnamon oil in vitro. Cinnamon oil reduced AHL production, diminishing biofilm development, swarming mobility, and synthesis of PCN, alginate, and proteases in PAO1. Cinnamaldehyde at sub-MIC levels has been shown to reduce swarming mobility and destabilize pseudomonal biofilms via reducing intracellular cyclic di-GMP [190]. Combining cinnamaldehyde with

tobramycin effectively inhibited QS, PAO1 biofilm formation, and dispersal of already existing biofilm [191].

The combined effects of cinnamaldehyde and gentamycin were advantageous and might potentially treat QS. The combined treatment effectively inhibited the production of AHL and reduced the level of expression of key QS genes in *P. aeruginosa* PAO1 [192]. The molecular docking study showed a strong association between cinnamaldehyde and QS receptors, indicating its capability as a QS modulator. Cinnamaldehyde effectively eliminated already-present biofilms and inhibited the formation of new biofilms by inhibiting EPS synthesis. Moreover, cinnamaldehyde effectively hinders *P. aeruginosa*'s QS circuitry by lowering QS gene expression and AHL synthesis [193].

Cinnamaldehyde's functionality is due to its strong association with *P. aeruginosa*'s QS receptors and inhibits the action of virulent factors and biofilm formation in the QS system, resulting in decreased pathogenicity of the bacteria. Kart et al. [194] found that combining cinnamaldehyde and ciprofloxacin resulted in a higher reduction in the minimal biofilm removal level than ciprofloxacin. Cinnamaldehyde was observed to impede QS and alginate production, inhibit PAO1 from forming biofilms and improve ciprofloxacin's efficacy against biofilms. Remarkably, cinnamaldehyde-embedded chitosan NPs at sub-MIC concentrations were fabricated to impede QS and *P. aeruginosa* biofilm development [195].

Compared to cinnamaldehyde, nanoformulations inhibited PAO1 swimming and swarming mobilities, biofilm, EPS, LasA protease, PCN, and RLs synthesis. This bioactive's high QQ activities make it a potential breakthrough in combating this infectious condition.

3.3. Terpene and terpenoids

Terpenes are naturally occurring chemicals with diverse structures derived from isoprene subunits, as shown in Fig. 4. Essential oils, di and

triterpenoids include monoterpenes, which have antibacterial and anti-biofilm effects and have been traditionally utilized as natural remedies [196]. Terpinen-4-ol (Fig. 2.f), the major biological active ingredient of tea tree oil, inhibited QS at sub-MIC levels and decreased the transcription of QS genes (*lasI*, *lasR*, *rhlI*, *rhlR*, *rhlAB*, *lasB*, *aprA*, *toxA*, and *plcH*) [197]. β -sitosterol, α -amyrin, and cassipourol—three terpenoids derived from *Platostoma rotundifolium*—have demonstrated anti-QS characteristics and inhibition of *las* and *rhl* systems against *P. aeruginosa*. Although every reported terpenoid could stop biofilm creation, only cassipourol and β -sitosterol could change the expression of genes linked to QS in *P. aeruginosa* [198].

Diterpene phytol, a biological compound, effectively inhibits biofilm development, twitching, and flagella movement with a 0.5 μ g/ml sub-MIC in *P. aeruginosa*. *Dysidea avara*, a Mediterranean sponge, and its secondary metabolic component, sesquiterpene hydroquinone avarol, demonstrated mild anti-QS effects in *P. aeruginosa*, reducing PAO1 biofilm development by 75 %, PCN synthesis by 39 %, and hindering twitching and protrusion capability of movement [199]. Subsequent examinations revealed that medicated *P. aeruginosa* PAO1 had fewer factors associated with virulence, supporting the findings of QS transcription investigations. Furthermore, terpinen-4-ol and ciprofloxacin functioned together to enhance the antibiotic's effectiveness towards *P. aeruginosa*.

This characteristic renders this biogenic substance extremely noteworthy since it suggests that using both old medications in conjunction with a novel antimicrobial that can boost or reinstate their efficiency would be an appropriate action approach for ensuring the continued efficacy of vital antibiotics.

3.3.1. α -Terpineol

α -Terpineol (Fig. 2.g) is a terpenoid alcohol widely used in the fragrance sector and is a crucial part of essential oils produced from tea

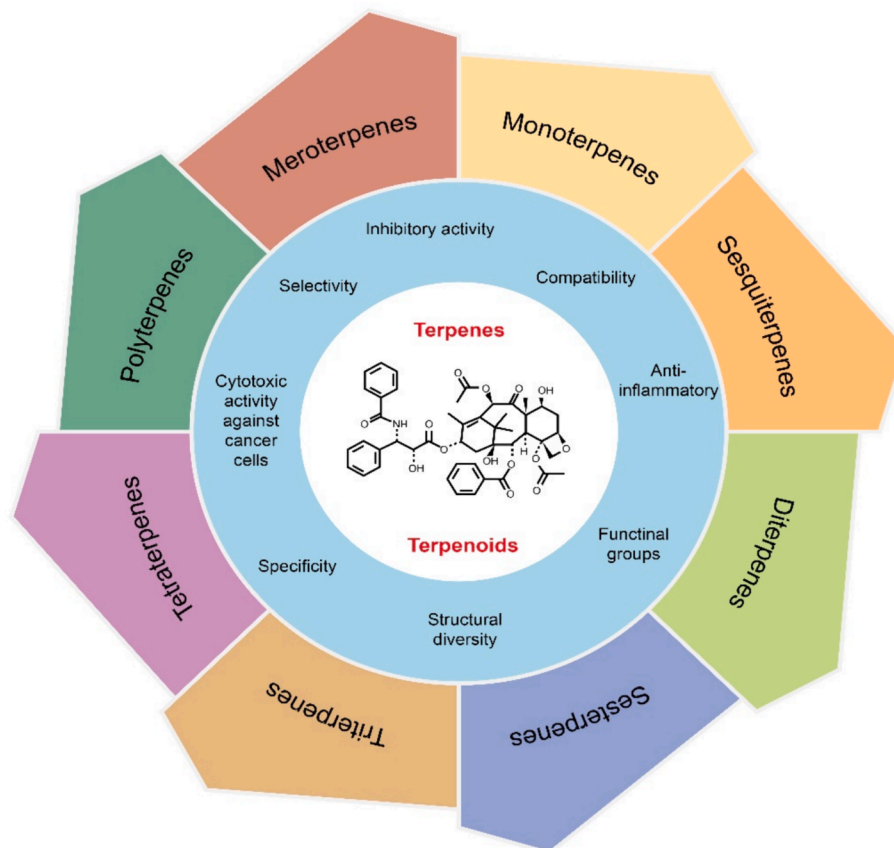


Fig. 4. Classification of terpene and terpenoids and their classification.

trees, pine trees, and other plants. Bose et al. [200] demonstrated the QQ activity of terpineol-containing nanostructured lipid carriers (NLCs) against *P. aeruginosa*. After receiving α -terpineol therapy, the transcription levels of several important QS genes (*las*, *rhl*) and pathogenic coding genes (*rhlAB*, *aprA*, *lasB*, *toxA*, and *plcH*) were substantially decreased. Significant reductions were seen in swimming mobility, the creation of biofilm, EPS, hemolysin, elastase, and PCN production.

The substantial binding potential of α -terpineol against the QS receptors, LasR, RhlR, and PqsR was also indicated by in silico investigation, which might explain its extensive activities in vitro. It is fascinating to note that administering nanolipoidal α -terpineol toward *P. aeruginosa* exhibited an anti-biofilm impact and triggered immune-modulating responses by raising the levels of inflammatory cytokines, such as IL-2, TNF- α , and macrophage-inhibiting protein-2, in the affected eyes [201].

3.3.2. Terpinen-4-ol

The isomeric form of α -terpineol, terpinen-4-ol, is a natural existing monoterpene serving as tea tree oil's primary biologically active component. It has been demonstrated that thyme essential oil, which contains various plant chemicals, especially terpinen-4-ol, suppresses QS, which lowers *P. aeruginosa*'s flagella-independent mode of movement and PCN synthesis by 74 % [202]. Noumi et al. [203] investigated terpinen-4-ol's QQ capability on PAO1 swarming mobility. The PAO1 strain's migration ability was significantly reduced by 25 % when terpinen-4-ol was introduced. A recent study found that terpinen-4-ol can reduce infectiousness, biofilm development, and QS mechanisms in *P. aeruginosa* [197]. Terpinen-4-ol showed a strong association with LasR, RhlR, and PqsR QS receptors when investigated in silico. Terpinen-4-ol at sub-inhibition levels downregulating the levels of transcription of *lasI*, *lasR*, *rhlI*, *rhlR*, *rhlAB*, *lasB*, *aprA*, *toxA*, and *plcH* QS in the *P. aeruginosa* PAO1 strain. Disrupting biofilm morphology resulted in a weakened biofilm with lower bacterial adhesion.

3.3.3. Citral

Approximately 70 to 80 % of lemongrass oil is composed of the terpenoid citral, which gives the plant its distinct flavor and scent. Citral (Fig. 2.h), at sub-MIC (840 $\mu\text{g/ml}$), has been demonstrated to have a 93 % inhibitory effect on *P. aeruginosa* swarming mobility and a notable decrease in PVD synthesis. Several citral derivatives (CD1, CD2, and CD3) have been constructed and studied in vitro to improve QS and biofilm development effectiveness.

It may be used as a prototype for conducting tests on *P. aeruginosa* pathogenic factors and QS mechanisms [204]. A research effort aimed to improve citral's ability to dissolve and QQ capability by converting it into a glycomonoterpene by glucose conjugation process [205]. The conjugate was demonstrated to eliminate QS by suppressing the synthesis of violacein, PVD, and *pseudomonas* biofilms at much lower doses than citral alone.

3.3.4. Linalool

Linalool accounts for around 70 % of the volatile concentrations for certain essential oils derived from fragrant plants [206]. In a recent study, linalool derived from *Ocimum tenuiflorum* plant extracts inhibited the synthesis of PCN, elastase and protease in *P. aeruginosa* [149]. Glycolinalool functioned as QSI towards establishing biofilm in *P. aeruginosa* NCIM 5029 [207].

3.3.5. Parthenolide

Tanacetum parthenium, a plant noted for its therapeutic qualities, yields parthenolide, a sesquiterpene lactone derived from the plant's active ingredients, sesquiterpenes and sesquiterpene lactones [208]. The sesquiterpene lactone found in feverfew plants may inhibit the generation of pathogenic factors by *P. aeruginosa* PAO1. Parthenolide (Fig. 2.p) at sub-MICs significantly reduced biofilm buildup while not impacting the development of bacteria. According to the reverse transcription polymerase chain reaction (RT-PCR) technique, downregulating of AI

synthases (*lasI* and *rhlI*) and their receptors (*LasR* and *RhlR*) resulted in reduced levels of various virulent factors, including PCN, protease, and swarming mobilities in *P. aeruginosa* [209]. When parthenolide was administered at a sub-MIC dose of 1 mM, there was a substantial reduction in virulent factors and biofilm development.

Dendrobium chrysotoxum contains parthenolide, which inhibits cellular adhesion, QS factor and fibronectin attachment by focusing on the enzyme SrtA or downregulating staphylococcal surface protein A (SpA). It also blocks infectious factors associated with *P. aeruginosa* and inhibits biofilm development [210].

3.3.6. Geraniol

Geraniol is a monoterpene identified in various vital oils of fruits, vegetables, and aromatic plants comprising citronella, lavender, lemongrass, and rose oil and is widely explored for therapeutic activities [211]. Li et al. [212] discovered that geraniol could suppress three QS mechanisms in *P. aeruginosa*, involving *las*, *rhl*, and *pqs*, by reducing the expression of essential QS genes. Further, QS-regulated phenotypes such as biofilm, PCN and elastase were also inhibited.

3.3.7. Carvacrol

Carvacrol (Fig. 2.n), a phenol monoterpene, is commonly present in essential oils extracted from aromatic plants such as pepperwort (*Lepidium flavum*), oregano (*Origanum vulgare*), and thyme (*Thymus vulgaris*) [213]. Carvacrol possesses biological characteristics, including cancer prevention, antioxidant, and antimicrobial properties [214,215]. Recent investigations found that carvacrol effectively inhibits the biofilm of *P. aeruginosa* [216,217], as shown in Fig. 5. According to Walczak et al. [218], carvacrol hindered the build-up of biofilms by 88 % for *P. aeruginosa* and diminished the enzymatic action of the biofilm by 40 %–100 %. Rodriguez et al. [219] reported that carvacrol at 0.9–7.9 mM reduced *P. aeruginosa* biofilm development on steel substrates by 1.5–3 log CFU cm^{-2} . Carvacrol at 0.7 and 3.9 mM inhibited violacein and PCN synthesis by around 50 and 60 % rather than impacting the proliferation of bacteria or survivability. Recent research has examined carvacrol's anti-QS potential toward *P. aeruginosa* using in vitro tests and docking investigations [220].

Carvacrol reduced the formation of C4-HSL (63 %), C6-HSL (80 %), and C12-HSL (60 %) in a concentration-specific manner, with sub-MIC level of 1.9 mM. It has been observed that carvacrol fully inhibits the expression of the *lasR* gene while leaving the *lasI* gene unaffected. As a result, QS-triggered pathogenic factors such as swarming mobility, biofilm creation, PCN synthesis, LasA protease, and LasB elastase were significantly reduced. Gobin et al. [221] demonstrated that the synergistic effect of gallic acid and carvacrol eliminated established

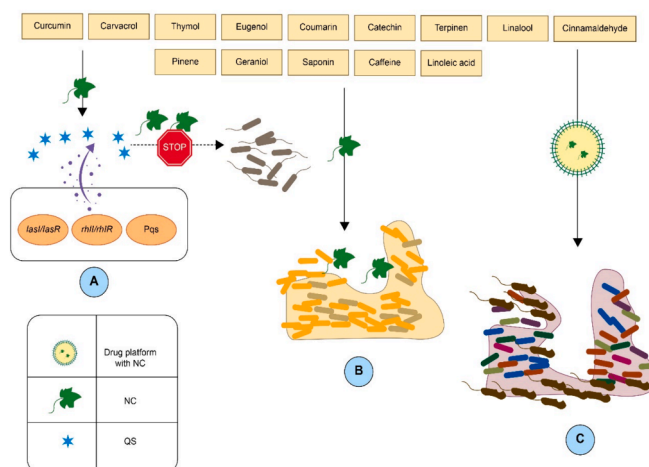


Fig. 5. Associations between plant-based natural compounds and *P. aeruginosa* biofilm.

P. aeruginosa biofilms. Gallic acid's ability to combat biofilms effectively may be related to modifications in the properties of the plasma membrane. Gallic acid promotes the release of K^+ from *P. aeruginosa*.

Intracellular components, such as ions, can cause cell death by altering gene expression, disrupting ion homeostasis, and interfering with signaling pathways. Gallic acid's anti-adhering activity disrupts biofilm, increasing the efficiency of carvacrol. It leads to bacterial membrane permeabilization and mortality [222,223]. Furthermore, carvacrol's penetration into *P. aeruginosa*'s biofilm community may be facilitated by chitosan NPs. Consequently, the ability of drug systems to effectively increase the antibiofilm action of carvacrol indicates the potential use of these scaffolds as hydrophobic chemical or systemic drug delivery in addition to therapeutic agents [224–226].

Phase (A) involves suppressing the QS system to prevent the plant-derived natural compound's initial stage of biofilm development. In phase (B), natural compounds further destroy biofilm communities, and phase C involves employing several drug mediums that can improve the natural compound's antibiofilm effectiveness. NC (natural compounds).

3.3.8. Thymol

Thymol is a phenolic monoterpene with a strong scent that is soluble well in alcohol and other natural solvents but weakly in water. Thymol is referred to as 2-isopropyl-5-methyl phenol [227]. Thymol, found in some plants, has long been used in conventional healthcare due to its medicinal properties [228]. Thymol is known to have antibacterial and anti-inflammatory effects on microorganisms. This compound's strong antibacterial properties make it beneficial for curing airway infections, dental illnesses, and gastrointestinal issues [229].

Thymol's antibiofilm action could derive from its capacity to break drug-resistant bacteria's surface adhesion and harm cellular membrane architecture toward *P. aeruginosa* [230]. Thymol's antibiofilm characteristics revealed that at a sub-MIC level of 25 mg/mL, this potent plant extract demonstrated the highest biofilm dispersion capability (at an efficiency of 80 %) and biofilm inhibiting impact (at a percentage of over 66 %) towards *P. aeruginosa* [231]. *Plectranthus amboinicus* leaf extractions demonstrated bactericidal and antibiofilm properties, reduced *P. aeruginosa*'s mobility and hydrophobic nature and were vital factors in the pathogen's growth [232]. Thymol, obtained from the root exudates of *Sedum alfredii*, can prevent *P. aeruginosa* from producing protease and elastase. It is accomplished by decreasing *lasB* expression, which has no noticeable effect on the main *las* system. Thymol appears to be a promising bioactive QS inhibitor [233]. Velázquez-Carriles et al. [234] created a new nanohybrid architecture using zinc-layered hydroxide salt (ZnLHS) and thymol to limit the development of bacteria. The derived pharmacologic substrate significantly reduced *P. aeruginosa* biofilm by over 90 %.

3.4. Organosulfur compounds

The sulfur-containing chemical molecules known as organosulfur compounds (OSC) have been linked to the strong smells of allium-containing vegetables like garlic and onions. Additionally, they are widely available in cruciferous vegetables like broccoli and cabbage [235]. Garlic's OSC has various bactericidal characteristics, notably the ability to combat MDR strains of bacteria and shows antibacterial, antibiofilm, antitoxin medications, and anti-QS action. Garlic's primary OSCs are ajoene, allicin, and aliphatic sulfides [236].

3.4.1. Allicin

Garlic contains antioxidant-rich compounds, anti-inflammatory and antibacterial properties. Garlic consists of the beneficial phytochemical allicin (diallylthiosulfinate), popularly known as the 'heart of garlic' [237]. Allicin (Fig. 2.d), derived from garlic, inhibited the gene *rhl* associated with C4-HSL production. Additionally, it prevents PQS molecule formation in *P. aeruginosa* and hinders the gene expression levels linked to RLs synthesis, such as *rhlA*, *rhlB*, and *rhlC* [85]. Lihua

et al. [238] employed standard assessments and luminescence microscopy to evaluate the influence of allicin on QS-regulated virulence-related variables and biofilm development in *P. aeruginosa*. The study found that allicin at 128 µg/ml prevented bacterial attachment for up to 9 h, followed by the treatment and lowered EPS generation by trice in a dose-depending manner. Additionally, the biofilm structure changed to take on a more loosely and thin form. Allicin at sub-MICs fully inhibited the formation of PCN and RLs, whereas exotoxin synthesis was downregulated three times and elastase by ten times, respectively. It was discovered that allicin suppressed the formation of RLs (*rhlA*, *rhlB*, and *rhlC* genes), PCN (*phzB2* and *phzM* genes), PVD, and elastase in a concentration-dependent way. qRT-PCR analysis revealed significant suppression of *rhl*, *pqsA*, *pqsD*, and *pqsR* transcripts instead of impacting the *las* system. These have demonstrated the ethnopharmacological significance of allicin as a therapeutical agent that may neutralize *P. aeruginosa*'s QS-dependent virulent factors.

3.4.2. Ajoene

Ajoene (Fig. 2.b), an OSC bioactive from *Allium sativum* (garlic), has been widely investigated for suppressing QS and QS-related virulence-associated variables [239]. Harjai et al. [240] found that garlic extracts dramatically inhibited the production of QS signals and the pathogenicity mechanism in *P. aeruginosa* in vitro.

P. aeruginosa-triggered renal-infected mice receiving oral garlic extracts showed reduced renal bacterial numbers, and their kidneys were shielded from any potential tissue degradation. Jakobsen et al. [241] reported that ajoene inhibited the expression level of important QS-controlled genes, including *lasA*, *chiC*, *lecA*, *rhlA*, *rhlB*, and *prpL*, in a dose-dependent manner.

Furthermore, reporter studies were employed to investigate ajoene's anti-QS potential. Ajoene at 100 µg/ml reduced biofilm development in *P. aeruginosa* by depleting RLs, eliminating lytic necrosis of polymorphonuclear leukocytes (PMNLs). In a mouse lung infection model, ajoene at 12.5 µg/g removed bacterial cells more effectively than untreated murine. Ajoene-ciprofloxacin combination treatment was studied for its effectiveness towards biofilm-based mice with acute renal infection and *pseudomonas* biofilms [242].

In vitro conditions, this combined treatment suppresses biofilm development, swimming and swarming locomotion in *P. aeruginosa*. Compared to infected controls and solitary medication therapies, the combinational transfer considerably decreased the prevalence of bacteria in treated mice's kidney tissue. The nanoformulation of ajoene, a QS inhibitory agent, was created utilizing chitosan and dextran sulfate. The QSI-overloaded NPs (686 nm) exhibited a high zeta potential 38. The nanoformulation was created to target *P. aeruginosa* biofilm-dependent mice kidney infection directly. The NPs and ciprofloxacin demonstrated considerable antivirulence efficacy in *P. aeruginosa*, lowering factors associated with virulence and biofilm.

The nanoformulation demonstrated two times higher antivirulence efficacy compared to ajoene itself. It greatly decreased virulent factors such as alginate, elastase, hemolysin, protease, pyochelin (Pch), and RLs [243]. Substantial biofilm-inhibiting efficacy has been connected to improved dissolution, accelerated absorption, and prolonged release of ajoene. Along with ciprofloxacin, pathogenic factor synthesis decreased by 73 %-97 %. Additionally, combining NPs and ciprofloxacin could eradicate biofilm above 99 %.

The nanoformulation contains chitosans, which can reduce the polarization of the cells of the cellular membranes that form biofilm and improve its accessibility to NPs. In vivo, monitoring employing fluorescent tagging validated the targeted administration of NPs. Bacteriological and histological tests validated the curative potential of the nanoformulation of ajoene [244]. In addition, combining ajoene with tobramycin increases the antibiotic's antibacterial properties by 80 %. This effect is feasible since QS suppression hinders the proper production of biofilms, making bacteria more susceptible to the effects of drugs and the immune system's responses [245].

Ajoene has been shown to suppress QS through alternative molecular pathways. Ajoene has been demonstrated to impede small regulatory RNA (sRNA) in *P. aeruginosa*, suppressing QS [246]. Ajoene was discovered to significantly suppress the expression of RsmY and RsmZ sRNAs within the initial static phase of bacterial development in a dose-responsive manner utilizing reporter assays and qRT-PCR. Ajoene architectural equivalents have also received attention lately for their strong QS process inhibitory properties. These ajoene equivalents have effectively targeted QS action, inhibiting virulence factors, such as in vitro synthesis of elastase, PCN, and RLs [247]. Additionally, in a mouse infectious implant paradigm, the compounds were demonstrated to hinder and impair *P. aeruginosa*'s ability to build biofilms on implants. Plant bioactives have been shown to suppress distinct QS systems, reducing QS-controlled pathogenic factors in *P. aeruginosa*, as shown in Fig. 6.

4. Advantages and shortcomings of utilizing QQ phytochemicals

The prevalence of bacterial illnesses is still rising amid the development of new-era antibiotics, which constantly threatens the world medical community [276]. These QQ compounds are thought to have the potential to exert selective pressure, which might lead to the emergence of drug resistance. There is limited data concerning this issue, but it is important to realize that no therapy technique is completely effective [277,278]. Combining QQ phytochemicals with

antibiotics can improve the reuse of drugs by focusing on several bacterial signaling channels and lowering selective pressure. Recent investigations showed the combined effect of anti-virulent phytochemicals and drugs [279]. Furthermore, pathogenic bacteria can repeat their metabolic processes or choose polymorphisms that provide resistance to QSIs.

It has been demonstrated that clinical isolates of *P. aeruginosa* derived from individuals with cystic fibrosis exhibit immunity to QSIs like brominated furanone by causing genetic changes that improve bacteria efflux capability [280]. Therefore, infectious bacteria can build defenses that interfere with QQ through several means. Remarkably, investigations have shown *P. aeruginosa* QS mutants resistant to the QSSMs. However, these mutations may confuse QSIs, diminishing their therapeutic relevance and effectiveness [281]. Pseudomonal pathogenesis involves three interrelated QS pathways, rendering it difficult to focus on with QQ phytochemicals. The existence of numerous QS mechanisms can lead to heterodimers of transcription controllers randomly interacting with various promoters, resulting in variation in gene expression in bacterium to avoid QSIs [282]. The pseudomonal Pqs system modulates pathogenic drivers, leading to cell death and the dispersal of eDNA and thus provides the infectious agent with survivability benefits [283]. Additionally, infectious bacteria can alter their QS receptors to improve affinity by changing the QSSM-adhering pocket's amino acid residuals [284].

QSIs with high affinities for QS receptors can effectively reduce

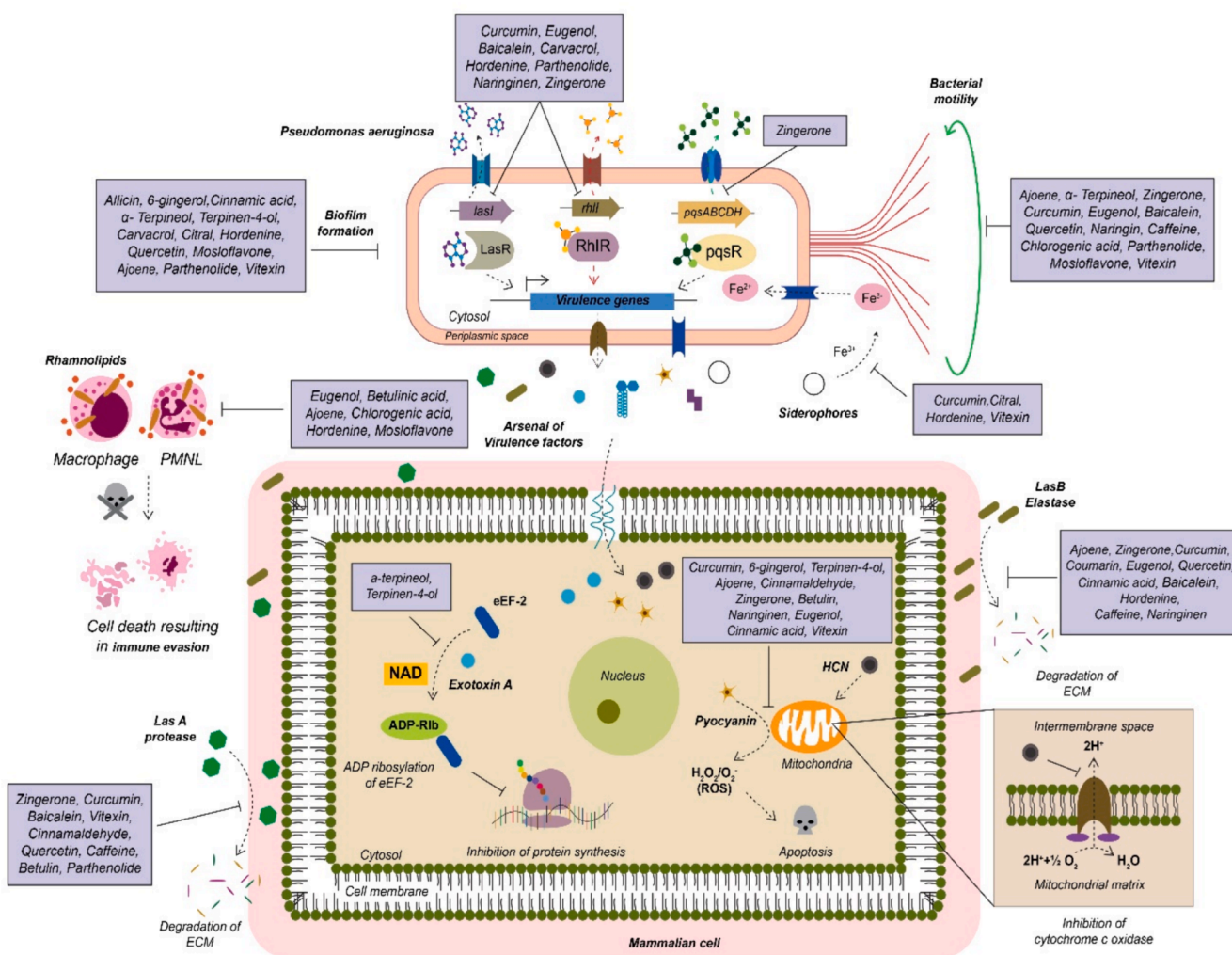


Fig. 6. Mechanistic illustration of phytochemicals that inhibit QS and target pathogenesis in various phases and *P. aeruginosa* QS-controlled virulent drivers that alter the host's physiology.

pathogen pathogenicity. Biogenic anti-QS phytochemicals may have limitations, but chemically improved those with appropriate functionalized groups can enhance associations. Biologically active phytochemicals are essential in regulating the host immune mechanism by promoting humoral and cell-driven reactions, providing anti-oxidants, preventing responses to inflammation and autoimmune illnesses, and preserving a balanced gut microbiota [285,286]. Even though phytochemicals provide several advantages, their impact on the variety, surplus, and balance of the mammalian microbiota has not been thoroughly studied. The impact of the human microbiome on illnesses, such as malignancy, highlights its importance in influencing the health of individuals [287].

QQ phytochemicals can improve human wellness by influencing the microbiota of several organs. Plant-based chemicals have difficulties in antivirulence treatment, including cytotoxicity, inadequate water dissolution, and low absorption. However, they may be managed for the benefit of humanity by developing efficient administration methods with a high therapy index [288]. As a result, the following characteristics of a perfect QQ phytochemical may be highlighted as follows: (i) reduced molecular weight; (ii) strong affinity for QS mechanism without causing any harm or deviating from the intended infection in the bacterium; (iii) sufficient accessibility and strong stability; (iv) resisting to degradation of enzymes by the host system; (v) minimalist or no cellular toxic effects for the host, making it suitable for human usage and (vi) there are no adverse impacts on the host microbiota [289,290]. Natural QSIs require further clinical research to ensure their long-term effectiveness and capacity to overcome antimicrobial resistivity. Potent antivirulent medications will soon become part of healthcare settings. Expanding knowledge on QSI, especially QQ phytochemicals, their possible toxic effects, and therapeutic potential in health care is encouraged.

Collaboration between drug companies and academic institutions can bridge the gap in reusing phytochemicals with bioactive properties for anti-virulence treatment [291,292].

5. Conclusion and future outlook

The excessive use of drugs in medical facilities has led to the rise of *P. aeruginosa*, an MDR bacterium that poses a significant danger to public healthcare settings. This pathogen's QS modulates pathogenic factor release and biofilm development, which is necessary for microbial adaptability to environmental alterations. As a result, perhaps one of the more intriguing ways to reduce *P. aeruginosa* virulence is targeting its communication mechanism. Thus, alleviating the seriousness of infections in clinical settings may require focusing on the QS mechanism that controls pathogenesis in *P. aeruginosa*.

Bioactive phytochemicals derived from plant kinds have proven beneficial in this specific scenario. Natural anti-virulence herbal remedies offer new opportunities for scientists to create therapies targeting *P. aeruginosa* by disrupting its QS systems and reducing its ability to transmit disease. In vitro and in vivo investigations have identified secondary metabolic compounds with anti-QS and antibiofilm properties that impact *P. aeruginosa* pathogenicity. Nevertheless, additional research is needed for effective implementation in clinical settings to prevent pseudomonal illnesses. Instead of developing novel antibiotics, research experiments should focus on identifying and repurposing anti-pathogenic naturally occurring chemicals, bioactive molecules, and synthetic medicines. Proteomics provides advanced tools for investigating protein-small molecule associations, which is critical for understanding the mechanisms that act and repurpose bioactive compounds as QS inhibitors in *P. aeruginosa*. Key strategies include (i) Affinity Purification-Mass Spectrometry (AP-MS), where bioactive compounds with affinity labeling are utilized to isolate interfering proteins from *P. aeruginosa* lysate. The protein combinations are subsequently determined using mass spectrometry (MS).

AP-MS can detect direct associations among bioactive compounds

and QS-associated proteins, revealing main action pathways and displaying feasible unforeseen interactions between proteins that may affect their repurpose as QS inhibitors. (ii) Thermal Proteome Profiling (TPP), where thermal gradients are used to determine the stability of protein of *P. aeruginosa* cells that are subjected to treatment with bioactive compounds. Stability alterations are indicative of protein associations with the biologically active molecule. TPP maps protein targets of biofunctional compounds, offering an understanding of their mechanisms of action and subsequent effects, which is critical for determining their selectivity and effectiveness as QS inhibitors [293].

(iii) Chemical proteomics, where bioactive compounds amended with reactive motifs that establish covalent interactions with targeted proteins. These complexes are extracted after cell death and examined. This strategy systematically uncovers bioactive compounds' binding locations on QS-related proteins, revealing comprehensive processes of action and off-target associations crucial to their repurpose as QS inhibitors. (iv) Label-free quantitative proteomics analyses variations in abundances of proteins and post-translational modifications (PTMs) to identify impacted systems and secondary effects essential for repurposing as QS inhibitors and (v) Activity-Based Protein Profiling (ABPP) specifically tags functional enzymes, detecting those suppressed by bioactive chemicals and providing useful information. ABPP detects functioning enzymes hindered by bioactive compounds, revealing operational insights into their modes of action and possible off-target enzyme inhibition, which is critical for repurposing these chemicals as QS inhibitors. Incorporating proteomics into the investigation of bioactive molecules in *P. aeruginosa* is critical for finding efficient QS inhibitors and exploiting preexisting bioactive chemicals for potential anti-virulent applications [294].

Compliant nanocarriers can also be developed to improve administration and accessibility. Compounds can be modified chemically or structurally to enhance adherence to potential therapeutic targets in bacterium QS systems. Furthermore, a wide range of clinical investigations on the impact of QSIs on infectious disorders should be prioritized. It will provide tangible and credible information before the anti-virulence technique is utilized for humanity's betterment. It is important to study how anti-virulence medications might cause human resistance to infectious bacteria to prevent resistance to QSI. Nano-level materials can be coated with biologically active phytochemicals to enhance their curative potential to combat nosocomial illnesses caused by pseudomonal biofilms on healthcare equipment. Composite nano-based coatings with phytochemicals or QQ enzymes, like acylases, lactonase and paraoxonase, can provide comprehensive antibiofilm activity on healthcare equipment. These improvements can help reduce the worldwide dependency on drugs. Implementing these strategies can minimize medication resistance and the healthcare expenses caused by *P. aeruginosa*, resulting in fewer hospitalizations and improved patient health outcomes. It will deliver significant social and economic advantages, enhancing today's life and future generations.

CRedit authorship contribution statement

Kayeen Vadakkan: Writing – original draft, Visualization, Investigation, Conceptualization. **Kuppasamy Sathishkumar:** Writing – review & editing, Visualization, Investigation. **Viji Ouseph Mapranathukaran:** Supervision, Investigation, Conceptualization. **Ajit Kumar Ngangbam:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Bijayalakshmi Devi Nongmaithem:** Writing – review & editing, Supervision, Data curation. **Janarthanam Hemapriya:** Conceptualization, Data curation, Investigation. **Jyotsna B. Nair:** Supervision, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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